

Node.js & Express Bootcamp – Module 4 Exercise

Hands-on Exercise No. 4

Course: Node.js & Express Bootcamp

Module: Scaling & Error Handling

Total Marks: 10

Due Date: 28/05/2026

Instructions

Prepare your solution in MS Word.

Submit in .doc or .docx format.

Late or copied submissions will not be accepted.

Learning Outcome

After completing this exercise, you will be able to:

- Implement error handling in Express applications.
 - Use logging tools for monitoring.
 - Apply scaling strategies for Node.js apps.
 - Understand load balancing and clustering.
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Problem Statement

You are working as a Backend Engineer for a growing eCommerce platform. The client wants the backend system to handle errors gracefully and scale to support thousands of users. Your task is to implement error handling and demonstrate scaling strategies in Node.js.

Task Requirements

Create a new Express project.

- Implement centralized error handling middleware.
- Add logging using a package like winston or morgan.
- Simulate an error (e.g., invalid route or missing data) and show how it is logged and handled.
- Configure Node.js clustering using the built-in cluster module to run multiple worker processes.
- Demonstrate load balancing by sending multiple requests and showing distribution across workers.

Take Screenshots of

- Error handling middleware code.
- Logs generated by requests.
- Cluster setup and worker processes.
- Console output showing load balancing.

Submission Guidelines

Upload your .docx file with code snippets, screenshots, and explanations via LMS.